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Exploring land banking, perceived neighborhood conditions, and social capital: A case of Detroit, Michigan, USA

Yunmi Park [a,1], Jung-Eun Lee [b,2], Hyungchul Chung [c,3], Galen D. Newman [d,4], Tae-Hyoung Tommy Gim [e,*,5]

[a] Department of Civil and Environmental Engineering and Institute of Construction and Environmental Engineering, Seoul National University, Seoul, South Korea
[b] Department of Civil and Environmental Engineering, Seoul National University, Seoul, South Korea
[c] Department of Urban Planning and Design, Design School, Xi'an Jiaotong-Liverpool University, Suzhou, China
[d] Department of Landscape Architecture and Urban Planning, Texas A&M University, College Station, TX, USA
[e] Department of City Planning and Environmental Planning Institute, Seoul National University, Seoul, South Korea

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ABSTRACT

As shrinking cities proliferate, land banks have emerged as critical instruments for managing urban vacancy. This study examines the effects of land-banking programs on social capital (SC), mediated by perceived neighborhood conditions (PNC). Based on 598 survey responses from residents in Detroit, Michigan, the study finds that Land Bank programs affect SC through both direct and indirect pathways. Own-It-Now shows a statistically significant total negative effect on SC, as small direct positive influences were outweighed by larger negative indirect effects through PNC. Side Lot demonstrates a contradictory mechanism, with positive direct effects offset by stronger negative indirect effects, resulting in no significant total outcome. Demolition exhibits a direct negative effect on SC. These findings suggest that strengthening SC requires reinvestment and reoccupation strategies that not only promote occupancy but also generate tangible improvements in neighborhood conditions.

1. Introduction

There has been growing governmental attention on the increase in urban vacancies and property abandonment (Heim Lafrombois and Park, 2021). This increase is a widespread challenge, manifesting differently in various contexts: from declining industrial cities in the Global North, which often experience vacancy due to deindustrialization and population loss (Lee et al., 2023), to rapidly expanding urban centers in the Global South, where informal settlements and rapid, unplanned growth can also lead to underutilized or abandoned spaces (Roy, 2011). Irrespective of the specific context, the accumulation of abandoned structures can become both aesthetically unpleasing and unsafe, resulting in depreciation of adjacent property values, a diminished sense of community, or reduced private investments in the community (Díaz et al., 2011; Newman et al., 2018). Moreover, uncollected property taxes can place a considerable burden on the local government to maintain an adequate level of public service (Alm et al., 2014). Accordingly, the concept of land banking (LB) has gained traction globally as a strategic tool for managing vacant and underutilized public and private land. Although its origins trace back earlier (1960s), LB notably regained popularity, particularly in the United States, in the aftermath of the 2008 sub-prime mortgage crisis, which resulted in a significant increase in foreclosed properties that were left vacant (Sands and Skidmore, 2015).

The primary purpose of recently established LB in the US is to "deposit" vacant properties (VP), reoccupy them, and reintegrate them into the tax rolls through lease or sale (Tappendorf and Denzin, 2010). Additionally, LBs are responsible for maintaining, rehabilitating, or

* Corresponding author.
E-mail addresses: ym.park@snu.ac.kr (Y. Park), jelee823@snu.ac.kr (J.-E. Lee), hyungchul.chung@xjtlu.edu.cn (H. Chung), gnewman@tamu.edu (G.D. Newman), taehyoung.gim@snu.ac.kr (T.-H.T. Gim).
[1] ORCID: 0000–0002-7924–6518
[2] ORCID: 0000–0003-4084–4489
[3] ORCID: 0000–0002-7609–6112
[4] ORCID: 0000–0003-4277–5931
[5] ORCID: 0000–0002-6623–1042

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demolishing VPs to mitigate the negative effects of vacancy in the community while identifying entities to purchase those abandoned properties. Often, tax-delinquent properties can be allocated to the city for community benefit such as affordable housing, parks, or green spaces, in close collaboration with local governments and related organizations (Center For Community Progress, 2019). In recent years, LBs have been newly established or are evolving to take on new roles in many shrinking cities—defined not by short-term demographic fluctuations but by long-term (20 years or more) structural decline normally characterized by deindustrialization, population loss, and rising housing vacancies (Park and LaFrombois, 2019; Schilling and Logan, 2008; Wiechmann and Wolff, 2013)—to address the persistent issue of vacant properties (VPs). In response, a wide range of programs is being developed to support the diverse goals of land banks. For example, as of April 2023, there were 311 LBs in the United States in various locations at the city, county, and state level, with more than 71.7 % having been founded since 2010 (Center For Community Progress, 2024a).

As LB programs have been increasingly actively implemented in recent years, comprehensive research on the impacts of LB activities is also underway. Previous research and projects have reported or assumed the ecological, economic, and social value of the operation of LBs (Park et al., 2022; Small and Minner, 2024). Most of these studies, however, are limited to examining the positive, or, at times, negligible impacts on the economic side (Heckert, 2015; Niemesh et al., 2019; Whitaker and Fitzpatrick IV, 2016), primarily focusing on demolition, housing values, and ecological aspects (e.g., ecological restoration in urban areas, urban agriculture) (Haase, 2008; Kim, 2016a). Much of the literature explores the negative externalities related to VPs. Many studies suggest that LBs, as a mediation tool for VPs, can help mitigate many of the negative externalities related to excess VPs. While still limited, there is some research on the two aforementioned areas, such as ecological and economic impacts, but very little attention has been given to the effects of LBs on social capital (SC). Among the existing studies, most focus on specific aspects of LB programs, particularly the smallest component, such as the transformation of VPs into community gardens or public green spaces, emphasizing their potential to serve as social gathering, collective ownership, and community participation (Anderson and Minor, 2017; Firth et al., 2011). However, these studies predominantly adopt a descriptive and qualitative approach with a greater focus only on community-based reuse practices.

Given this fact, this study explores the direct and indirect impacts of various LB programs (i.e., demolition, Own-It-Now, Side Lot, and Partner Sales) and LB-owned properties on SC, measured by support and neighboring, place attachment, and community satisfaction. The research uses 598 survey responses collected from Detroit residents in the US as of 2021. We hypothesize that the improvement of neighborhood condition (e.g., safety, cleanness, housing value, and racial mix) by LB programs helps mediate the relationship between reoccupied, transformed, and maintained abandoned homes and SC. To test this hypothesis, Weighted Least Squares Mean and Variance (WLSMV) was employed as a method.

2. Literature review

2.1. Land banking and its impact

Modern LBs differ from earlier versions, which primarily focused on securing agricultural land (Veršinskas et al., 2022) or facilitating future growth (Brennan, 1980). Contemporary LBs prioritize addressing vacancy and abandonment issues. Unlike their earlier counterparts, these LBs are non-governmental or quasi-governmental entities endowed with governmental powers, duties, functions, and responsibilities for managing tax-foreclosed properties. ax reforms and state-level legal amendments have facilitated LBs in acquiring, maintaining, and repurposing abandoned properties (Leonard, 2010). For instance, Michigan's 2003 Land Bank Fast Track Act enabled LBs to acquire and manage tax-foreclosed properties. These authorities, formed through agreements with the Michigan Land Bank Authority, are put in place to return abandoned or blighted properties to productive use, promoting economic growth and community revitalization. LBs are required to notify stakeholders of tax-delinquent properties, but they can proceed with acquisition without stakeholder consent once the tax debt is resolved (Kelly, 2014). Also, LB-owned real estate could be exempted from property taxes for five years after the sale, and land banked properties in Michigan may be eligible for tax relief upon purchasing (Tappendorf and Denzin, 2010). Additionally, community development corporations and other transferees have the opportunity to purchase these properties at a discounted fair-market value under a deed that includes public purpose restrictions (Gasteyer et al., 2015). Eighteen states and Puerto Rico currently have Land-Bank Enabling Legislation as of 2024 (Center For Community Progress, 2024b). As of 2023, 311 LBs operate in the U.S., offering programs in maintenance, demolition, housing, and lot sales. Maintenance activities include tasks such as repairs, boarding-up, trash removal, and clean-up, alongside active demolition efforts. Additionally, LBs often sell acquired properties "as is" or after rehabilitation. In such cases, LBs sell or auction vacant and abandoned properties to rehabbers or developers at below-market prices, requiring buyers to demonstrate a clear commitment, sufficient funding, detailed plans, and the ability to rehabilitate the property (Bollwahn, 2019). Often, LBs encourage residents to purchase adjacent vacant lots for repurposing them, allowing them to expand their yards, create gardens, build garages, and more. For example, Chicago's Large Lot Program sells VPs at minimal prices with ownership conditions (Stern and Lester, 2021). LBs also sell VPs or lots to community partners to demolish, rehabilitate, or repurpose them into community gardens, green spaces, lot beautification projects, or affordable housing developments (Hamilton County Land Bank, 2023).

While the United States has been the most active in institutionalizing Land Banking strategies, the issue of VPs is a global phenomenon, prompting similar initiatives in other countries as well. For example, Japan has also implemented institutional approaches to tackle the issue of vacant housing, particularly through the development of regionally operated 'Akiya Banks' (literally, banks for unoccupied houses). Initiated by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) under the 2015 Vacant Houses Special Measures Act, this system is supported by local governments using various funding sources such as urban revitalization, housing, and community regeneration budgets (Dang, 2024). By April 2023, 1318 out of the nation's 1718 municipalities had launched their own Akiya Bank programs (Akiya Banks., 2024). The issue of VPs is also prevalent in the Global South. Unlike in the Global North, where such spaces are typically managed within formal urban planning frameworks, vacant spaces in many cities of the Global South are often informally occupied and constitute a vital part of the urban economy such as self-employment activities, urban agriculture, religious practices, or educational services (Skinner and Watson, 2017). However, this informality often gives rise to tensions between public authorities and space users, and places informal users in precarious and unstable conditions thereby underscoring the growing need to incorporate temporary uses into formal planning and regulatory systems to ensure equitable and secure access to urban land (Agheyisi, 2025). As such, while formal land banking institutions, like those in the United States, are uncommon in Latin America, many countries have implemented various fiscal and legal tools (e.g., property taxes to deter speculation, development incentives) and planning-oriented mechanisms (e.g., public-private partnerships for land reuse, social housing initiatives) to manage vacant land and encourage its productive reuse (de de de Araujo Larangeira, 2003).

The benefits of LBs can be categorized into three main domains: economic, ecological, and social. Economically, LBs help increase tax revenue, reduce maintenance costs, and stimulate neighborhood revitalization by promoting the repurposing of vacant homes and lots. The Philadelphia Redevelopment Authority (2010) declared that the value of properties located near vacant land decreased by 20 %, while the government annually invests $20 million to maintain VPs. Cook County's LB activities in 2023 generated $5.3 million in tax revenue, $13.8 million in private investment, and 144 job creations (Cook County Land Bank Authority, 2023). In addition, the LB produced a benefit-cost ratio of $3.23 for every $1 spent, with total spending amounting to $8.7 million. Previous studies that observed the impact of a few LB policies (e.g., simple acquisition, demolition, or rebuilding) on housing prices. In general, demolition showed positive or less negative impacts on nearby housing values (Hong, 2018; Niemesh et al., 2019; Whitaker and Fitzpatrick IV, 2016). Housing price appreciation linked to remodeling occurred exclusively in higher-income neighborhoods (vom Hofe et al., 2019). A recent study conducted in Detroit, U.S. examined a variety of LB programs over a 10-year period of housing transactions and found that their impacts differed by program. For instance, the Auction program led to an increase in adjacent housing values, whereas the Own-It-Now program showed no statistically significant effect (Lee et al., 2025). In addition, ecological and social impact are most often discussed in the context of community-level repurposing such as community gardens, open spaces, and public green spaces. Restoring vacant lands through re-greening, ecological rehabilitation, or the installation of green infrastructure—on vacant lots of recently demolished properties—could enhance ecosystem services, such as climate regulation, pollination, and recreation (Kim, 2016b). Many people also see the re-proliferation of grasses and trees following demolition as an excellent opportunity for environmental restoration (Haase, 2008). These initiatives are also believed to yield social benefits alongside their environmental impacts, as such spaces foster social interaction and encourage collaboration among residents across racial and generational lines (Schukoske, 1999).

The 2024 report from the Center for Community Progress highlights LBs' growing focus on social outcomes, beyond their initial emphasis on economic and ecological benefits (Graziani, 2024). Although the evidence is still quite limited, evidence from shrinking cities in the U.S. suggests that repurposing VPs can reduce crime. In Youngstown, Ohio, community-oriented reuse led to a notable decrease in crime density (Kondo et al., 2016), and in Columbus, Ohio, a rehabilitation and homeownership initiative was associated with reduced property-related crime (Kondo et al., 2022). Promoting affordability and advancing racial equity are also central goals of LB. Robinson and Woodin (2024) examined neighborhoods where the State Land Bank Authority of Michigan holds the most parcels and found that the LB owns a higher number of parcels in historically redlined neighborhoods. The authors suggested that if these properties are repurposed to accommodate new tenants, they could potentially yield positive impacts by reducing the concentration of socially vulnerable neighborhoods with minorities. It is well known that increased affordability promotes residential socio-economic integration and tenure security for less advantaged groups, such as the elderly and lower-income tenants (Mischiu, 2019). Moreover, by engaging in what is often described as 'voting with their feet,' residents who actively purchase and improve properties express a meaningful commitment to place (Wu and Tan, 2023). Rigolon et al. (2021) found that the empowerment of residents who seek social meaning in purchasing and remodeling properties plays a significant role in enhancing the greening of their own properties.

2.2. Social capital in declining neighborhoods

Human beings are inherently social creatures who exist within networks of relationships through which they exchange information, trust, and resources (Mead, 2015). It has long been recognized that these social connections significantly influence not only individual well-being but also group performance and broader social cohesion (Simmel, 2009). An early SC theorist, Bourdieu (1986) extended this notion by emphasizing that social networks can serve as structural conditions for generating resources and he positioned SC as a resource on par with economic, cultural, and symbolic capital. Building on this idea, Coleman (1988) identified various forms of SC, particularly emphasizing the role of family and community ties in forming SC. It was with Putnam (1995) the influential distinction between "bonding" and "bridging" SC was introduced; while bonding refers to strong ties within homogeneous groups that reinforce internal solidarity, bridging refers to weaker ties across diverse groups that promote social integration and civic cooperation. Since then, the concept of SC has broadened and evolved, encompassing multiple dimensions, levels, and interpretations across disciplines. For instance, Halpern (2005) viewed SC as consisting of different components (i.e., networks, norms, and sanctions), levels of analysis (i.e., micro, meso, and macro), and function (i.e., bonding [inward networking]), bridging (outward networking), and linking (asymmetric power relations). The same researchers also observed SC from different perspectives. Harpham (2008) proposed a conceptual framework that classifies SC into two distinct groups: structural (e.g., association links and networks) and cognitive (e.g., values and perceptions). Perkins and Long (2002) expanded the classification of SC into informal and formal aspects based on cognitive/trust and behavioral elements. Taken together, these evolutions and advancements imply that SC is not merely a collection of interpersonal connections; rather, it is best understood as assets embedded in trust, shared norms, and social networks that facilitate collective action to address common problems within communities—and at times, through connections that extend beyond them (Lang and Hornburg, 1998).

Parallel to the theoretical development of SC, substantial empirical evidence has emerged regarding its effects on both individuals and communities. For instance, previous studies found that SC is closely linked to overall quality of life by enhancing physical and mental health (Boyce et al., 2008; Mohnen et al., 2011), educational achievement (Caughy and O'campo, 2006; Woolley et al., 2008), walkability (Leyden, 2003; Rogers et al., 2011), and sense of security (Saegert and Winkel, 2004). Moreover, in communities with high levels of SC, residents tend to collaborate in pursuing common goals, adhere to local rules, and engage in collective decision-making (Adler and Kwon, 2002). SC becomes particularly important when it comes to declining neighborhoods. Residents in disadvantaged neighborhoods tended to rely more on reciprocal and informal support from their neighbors (e.g., repairing, child caring, and riding) because they often lack the resources needed (Schieman, 2005). Moreover, collective efficacy is likely to mediate criminal activities that occur more frequently in lower-income neighborhoods (Saegert and Winkel, 2004). It is evident that underrepresented ethnic enclaves, which likely have higher SC, actively respond to negative issues such as crime, poverty, and vandalism (Jones and Shen, 2014). This could be because a community functions as a cohesive social unit based on trust, feelings of togetherness, mutual support, and a sense of belonging. Moreover, a neighborhood with greater SC (e.g., socio-cultural milieu and institutional infrastructure) is less likely to succumb to the forces of decline and is more likely to remain stable over time (Temkin and Rohe, 1998). In that sense, it is crucial to examine whether policy interventions in declining neighborhoods can preserve existing SC while also fostering the creation of new SC.

2.3. Research gap & conceptual framework building

As discussed in the previous sections, LB is a project aimed at effectively addressing VPs in declining neighborhoods. While its potential social benefits have been acknowledged, empirical research on this aspect remains limited. Similarly, although some studies have investigated the negative effects of demolition (Kleinhans, 2009; Power, 2008, 2010) or the positive influence of community gardens (Glover et al., 2005; Kingsley and Townsend, 2006) on SC, these programs constitute only a part of the increasingly diverse approaches being adopted by. Or, although limited in number, most of the existing literature on LB and SC has relied on qualitative approaches, offering rich, in-depth narratives such as Kingsley and Townsend (2006). Therefore, quantitatively examining whether SC can be fostered through LB programs to revitalize these communities is crucial for advancing the literature and policy developments.

To address this gap, this study investigates how specific LB activities may shape SC, potentially through two key mechanisms (Fig. 1). First, the influx of new residents through property ownership and collective community activities facilitated by community-based repurposing projects can directly help restore and reactivate broken social ties and community dynamics. Moreover, by demonstrating their commitment through direct investment in the area, residents who actively purchase and improve properties express a meaningful commitment to place. To illustrate, if they are new homeowners or returned former residents (who were previously renters), they can reconnect with existing residents and quickly become a positive social force, further contributing to the revitalization of communities (Kelly, 2014). In addition, when long-term inhabitants leave, a neighborhood's social fabric is often prone to deteriorate. This deterioration, then, erodes community identity, community connection, social bonds, and affective responses (Meehan and Bryde, 2014; Rogers and Park, 2018). As such, supplementing community participation initiatives (e.g., community gardens and community-side projects) also devotes enhanced SC to a mutual sense of community responsibility and obligations toward the community, which results in the better formation of SC (Lelieveldt, 2004). Rigolon et al. (2021), in their study on Chicago's Large Lot program, emphasized that the remaining residents' decision to own property is driven not only by financial motivations but also emotional motivations that pertain to the community as a whole, often spanning multiple generations. In contrast, the dynamics of SC also shift when racially distinct groups move into a neighborhood, potentially posing threats to existing SC. Studies, for instance, have found that long-term Black residents in gentrifying neighborhoods, such as Central Harlem, were likely to face weakened social ties (Versey, 2018). The dynamics of SC also vary depending on who the incoming residents are. According to Butler and Robson (2001), they observed the newly arrived middle-class residents formed the strongest SC that led the change of neighborhoods, centered around a shared interest in education; the influx of high-income professionals and middle-class residents from diverse backgrounds had little impact on SC.

Second, we can assume that the indirect influence of SC is promoted through the physical improvement of VPs. Typically, ownership promotes the physical improvement of owned properties as well as adjacent public spaces. While not directly focused on LB, a comprehensive review by Wood et al. (2008) showed a significant link between the improvement of the neighborhood's physical condition (i.e., incivility and disorder, aesthetic appearance, housing design, private and public property upkeep, and greening) and SC. They argued that the condition of and level of care in a neighborhood (e.g., adequate lighting, meeting places, and private and public property upkeep) have an impact on community perceptions and the extent of social interaction. Stewart et al. (2019) also showed that beautification efforts can lead to increased social interaction among community members. Blight remediation efforts demonstrate community control and have a significant impact on safety perception and crime reduction (Hadavi et al., 2021; Stern and Lester, 2021), thereby fostering trust within the community (Tappendorf and Denzin, 2010).

Accordingly, LB programs are expected to potentially boost SC both directly and indirectly. However, the answers to these questions remain unclear. As such, this study attempts to explore two central questions:

Q1. Do LB programs have direct effects on SC?

Q2. Do neighborhood conditions mediate the effects of LB programs on the SC of the community?

3. Research design

3.1. Study area

Detroit's population in the U.S. peaked at approximately 1.85 million in 1950 but declined by more than 60 % by 2020, driven primarily by the decline of the automobile industry, rapid suburbanization, and the city's eventual fiscal collapse (Lee et al., 2025). This prolonged decline resulted in widespread vacancy, disinvestment, and urban blight across many neighborhoods, especially in the city's core. Now the city has the most vacant houses of any major city in the U.S. While there have been recent promising signs of improvement across the city, Detroit has lost 10.5 % of its population, and there has been an increase of 28.4 % in vacant houses over the past decade (Blanco, 2020; US Census Bureau, 2020). In response to these challenges, Detroit initiated the "Farm-A-Lot" program in the 1970s, which repurposed vacant land into small community gardens as an early strategy to manage urban decline (Bearre, 1976). These early efforts represented the city's first municipal-level responses to vacancy and blight. In 2009, the DLBA began to manage a significant portion of the city's vacant land, including many of the properties that were once part of the Farm-A-Lot program (Baker, 2020). In 2014, the DLBA's inventory expanded to include most of the city's publicly owned residential properties, consolidating efforts to manage these spaces effectively(Kwok, 2018). This was made possible in part by Michigan's legislative reforms, such as the Land Bank Fast Track Act, which streamlined title clearance, addressed tax issues, and enabled more flexible financing tools for land banks (Leonard, 2010). Since 2014, the Detroit Land Bank Authority (DLBA) has evolved into a large influencer on countering VPs in Detroit after being awarded a grant through the Neighborhood Stabilization Program by the Department of Housing and Urban Development. The DLBA used $2.9 million in 2013 to purchase and secure 30 foreclosed single-family and duplex properties (Detroit, 2021). The DLBA actively manages a vast majority of vacancies. The DLBA actively manages the vast majority of VPs in Detroit. While the 2020 American Community Survey (5-year estimate) reported a total of 87,966 vacant housing units (US Census Bureau, 2020), the DLBA's inventory included 22,061 vacant structures and 66,956 vacant lots as of April 2020 (Detroit Land Bank Authority, 2020). Although precise data on vacant lots in 2020 are not available, estimates from Motor City Mapping data set—based on observations collected by approximately 200 surveyors and associates—indicated the presence of

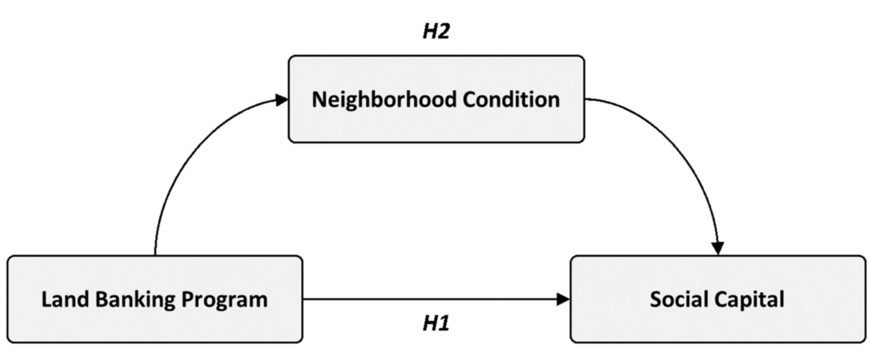

Fig. 1. Conceptual Framework.

about 74,313 vacant lots in 2014 (Nolan, 2017). While a direct numerical comparison is not methodologically appropriate due to definitional differences—particularly the fact that Census data include units that are only temporarily unoccupied—it is nonetheless clear that the DLBA serves as the primary entity responsible for managing VPs in the city. The DLBA have run several land-reuse and reoccupation programs such as (1) "Demolition" in cooperation with the city; (2) "Auction;" (3) "Own-It-Now" (properties sold as is at $1000); (4) "Rehab-&-Ready" (sold as remodeled); (5) "Partner Sales" (sold to community partners to enhance neighborhood quality through pocket and community gardens); (6) Side Lots (sold for $100 to residents who boarded at DLBA properties with fewer than 7500 ft^2); and (7) Neighborhood Lots (sold for $250 to residents who reside within 500 ft. from the DLBA properties). From 2014–2020, a total of 20,870 residential properties were demolished, and 15,006 side lots were sold, followed by 4677 Own-It-Now and 2666 Auction sales (Detroit, 2020). Considering the variety of its programs and accumulated cases, Detroit presents a good study area to demonstrate the impact of LB programs on SC.

3.2. Construct, variables, measurements, and data

Based on the conceptual framework in Fig. 1, constructs and measures were defined, and data were collected accordingly. The data on each LB-sold property type were retrieved through the City of Detroit Open Data Portal (https://data.detroitmi.gov/), which consistently updates the data on the sold properties in each LB program, while SC and other covariates were retrieved from the survey. The specific questionnaires for each construct are detailed in Table 1.

3.2.1. [Dependent variable: social capital]

As LB programs primarily operate at the neighborhood-level and are assumed to have a direct impact on neighborhood-level dynamics, SC was measured with a focus on bonding SC as conceptualized by Putnam (2000), which includes strong ties, mutual support, and a sense of shared identity among similar individuals within small and homogenous communities. For the operationalization the survey questions were derived from the previous studies: neighborly support, reflecting the readiness for mutual aid and reciprocity (Sampson et al., 1997); local social networks, indicative of dense, frequent interactions among residents (Curley, 2010; Perkins and Long, 2002); place attachment, capturing the emotional connection and shared belonging to the immediate physical environment (Anton and Lawrence, 2014; Brown and Raymond, 2007; Hesari et al., 2020); and community satisfaction, signifying collective contentment with the internal social fabric and sense of 'us' within the neighborhood (Kingston et al., 1999; Lochner et al., 1999; Parker et al., 2001). The internal consistency of the survey items was examined to assess the empirical support for measuring SC, yielding a Cronbach's alpha of 0.879, which indicates good reliability and scale consistency. These items were modeled as reflective indicators of a latent SC construct.

Table 1
Goodness of Fit.

Fit indices		Cutoffs (more conservative)	Our Model
$\chi 2$			474.872
	df.		179
	p	Insignificant	0.000
Relative $\chi 2$		≤ 3	2.653
RMSEA		≤ 0.08 (0.05)	0.053
	Low 90		0.047
	High 90		0.058
	p	Insignificant	0.220
SRMR		≥ 0.08 (0.05)	0.045
TLI		≥ 0.9	0.984
GFI		≥ 0.9	0.998
CFI		≥ 0.9	0.989

3.2.2. [Independent variable: LB programs]

The most prominent five LB programs were considered: LB-owned (properties remain vacant but are managed with no active actions taken), Demolition, Own-It-Now (sold as-is for as little as $1000), Side Lot (less than 7500 ft^2 and sold only to adjacent occupied homes $100), and Partner Sales (sold for community purposes). Then, the total number of properties involved in each program within 500 feet of the survey respondents' homes from January 2018 to December 2020 was counted, as a certain amount of time is required to observe noticeable and immediate impacts of land bank programs and their influence on SC.

While SC development often span longer periods (Woolcock, 1998), our three-year study period is appropriate for capturing immediate shifts in bonding SC at the neighborhood scale. At this scale, interactions are more frequent and direct, making the timeframe adequate to observe the impact of tangible physical changes and the resulting immediate shifts in how neighbors interact and perceive their community. Also, previous studies demonstrates that specific interventions and significant neighborhood-level changes can induce observable shifts in bonding SC within relatively shorter durations. For example, Kingsley and Townsend (2006) investigated SC formation in the "Dig In Community Garden" which started in 2002. Their interviews were likely conducted in 2004 or 2005, meaning they were observing SC development over a period of roughly 2–3 years after the garden's establishment. The relevance of observing short-term impacts is further underscored by the dynamics of policy interventions. Also, Ryu et al. (2018) conducted a survey on community-building projects completed in Seoul between November 2011 and December 2014, in order to assess changes in SC. The average project completion date was May 2013, and for 9 out of the 12 sites, the survey conducted in June 2015 took place approximately three years after project completion. As Koay and Dillon (2020) illustrate, the effects of policy interventions, such as housing demolition, can manifest as distinct short-term and long-term impacts (i.e., happiness of residents). Their research indicates that while the positive effect of demolition on happiness is noticeable if it happened recently (within the last three or four years), this effect can become small or even statistically insignificant if the demolition occurred in the past four to seven years. This time-dependent effect in intervention outcomes supports the notion that changes in social dynamics are not linear, underscoring the importance of observing their evolution over varying timeframes, including both short-term and long-term effects.

Moreover, since 2014—excluding the year when data collection began and the DLBA was established—on average of 7309 properties were sold annually through the five programs examined in this study. Sales volumes remained relatively stable across the study period. Notably, 2019 recorded the highest number of properties sold, with 9097 transactions. Even during the pandemic year of 2020 (=6453), sales remained comparable to levels observed in 2017 (=6911) and 2018 (=6819), indicating that the COVID-19 pandemic had no significant impact on the number of lots sold. Including the earlier years (2015–2017), which correspond to the initial phase of the program, may capture delayed effects, it also increases the risk of confounding influences and weakens causal attribution. Given this temporal distribution and supports from the previous studies, a three-year observation period following program implementation is considered appropriate for capturing immediate and obervable intervention effects.

This study does not rely on pre-defined administrative units (e.g., census tracts, administrative districts) as the unit of analysis. Instead, the spatial extent of each individual's perceived neighborhood is defined by creating a buffer of a specified radius around the respondent's residence. This approach is intended to more accurately capture the geographic area in which Land Bank (LB) program interventions are likely to have a meaningful impact. Spcifically, buffers with a 500 ft. radius were used in assuming that residents could see the immediate change around their walking or local viewing distance, as guided by previous similar methodologies such as Whitaker and Fitzpatrick IV (2016), Hong (2018), and Lee et al. (2025) that determined the impact of LB properties on housing value. Although the DLBA property count within the buffer serves as a valid proxy—since the DLBA manages the vast majority of VPs in Detroit, as noted in Section 3.1 Study Area—it is important to note that the effects of the land bank program may be over- or underestimated. This is due to the inability to control for nearby vacancy, or the repurposing and reoccupying of properties not managed by the DLBA, as a result of data limitations. Therefore, caution is warranted in interpreting the results.

3.2.3. [Independent variable: perceived neighborhood condition]

While neighborhood improvements through VP interventions are often associated with changes in various neighborhood factors such as safety, sanitation, aesthetics, and socioeconomic indicators as mentioned in the literature review section, objective data on neighborhood improvement in these dimensions is rarely available at the buffer-unit level, limiting the feasibility of incorporating such measures into the analysis. Therefore, this study measured perceived neighborhood condition (PNC) based on satisfaction levels with upkeep status, safety, overall improvement, cleanliness, and home value guided by previous studies (Dassopoulos et al., 2012; Lee et al., 2017; Mouratidis and Yiannakou, 2022), using a survey by means of a five-point Likert scale (1 = not at all satisfied to 5 = extremely *satisfied*). Each indicator was modeled as a reflective indicator of the PNC latent construct.

3.2.4. [Independent variable: other covariates]

To control for the impact of SC beyond neighborhood condition and LB programs, other possible correlates of SC were included based on the previous studies (Leviten-Reid and Matthew, 2018; Mullenbach et al., 2022): homeownership (owner vs. renter), length of residence (≥10 years= 1; otherwise=0), household income (<$25,000 = 1, otherwise=0), and age (≥40years=1, otherwise=0). Individual characteristics—except for homeownership—that were originally measured as categorical variables were transformed into dummy variables based on theoretically and empirically meaningful thresholds: 10 years or more can be considered long-term residence (Ghasri et al., 2023; Sergeant and Ekerdt, 2008), 2) $25,000 almost aligns with the federal poverty line of $26,500 for a family of four in 2021 (U.S. Department of Health and Human Services, 2021), and 3) residential mobility tends to be high before age 40, reflecting a short-term orientation toward residence, while after age 40, individuals become increasingly settlement-oriented and invest more in long-term community stability (Clark and Onaka, 1983; Reid, 2004). Having an empirically meaningful split point avoids imposing unjustified interval-level assumptions and allows the model to capture potential threshold or non-linear effects between the predictors and the latent outcomes. In the case of length of residence, various break points (5, 10, and 20 years) were tested, but the results remained robust regardless of the threshold applied. As the respondents had to reveal their residential address, other personal details were not asked to protect their privacy.

3.3. Survey procedure and description

An online survey was administered by Qualtrics from July 23 to September 27, 2021, following IRB approval (OOOO-202107–0017–01). Participants were restricted to current residents of Detroit as of 2021 by utilizing the following screening question: “Are you currently a resident of the city of Detroit?” Additionally, respondents were required to provide their residential addresses. Initially, the research team was provided 632 responses, but non-Detroit residents were filtered out through address verification (some addresses were missing, while others were located outside the Detroit boundary). After five rounds of collecting more responses and filtering between authors and Qualtrics, a total of 598 responses were retained for analysis (Fig. 2). Since all survey questions were mandatory, there were no missing data, as respondents were unable to proceed without answering each question.

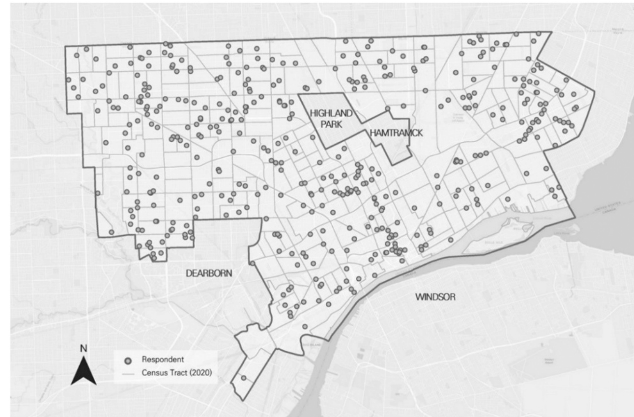

Fig. 2. Study Area and Location of Survey Respondents.

More than half of the respondents were African American (60.0 %) followed by White (35.6 %) and Asian (1.2 %), then or other (3.2 %). Approximately 52 % of the respondents were not homeowners, and 26 % had lived in their residence for more than ten years. These statistics does not completely match but is similar to Detroit’s, (2021) American Community Survey; approximately 76.6 % of Detroit’s residents identified African American and 52.6 % of the population were homeowners. This discrepancy may be attributed to the use of an online survey, despite employing a random sampling method.

3.4. Analysis

Given the ordinal nature of the Likert-scale indicators to construct SC and PNC and binary control variables, WLSMV estimation was employed instead of Maximum Likelihood (ML) (Flora and Curran, 2004).

Three data conditions were assessed to justify the use of WLSMV estimation. First, the sample size satisfies requirements for WLSMV estimation in SEM analysis, as recommended by Nussbeck et al. (2006). They suggested that WLSMV performs adequately with sample sizes of 200 or more for simple models, though larger samples (500 or greater) are preferred for more complex models. Given our model complexity with 97 free parameters—31 regression weights, 41 covariances, and 25 variances—our sample size of 598 is adequate for WLSMV estimation. Second, distributional properties of the observed variables were examined. Four out of 14 indicators slightly exceeded the 20 % floor/ceiling threshold suggested by Murugappan et al. (2022)—two indicators in PNC showed floor effects (*safety* = 25.9 %, *clean* = 21.1 %) and two in SC showed ceiling effects (*is_2* = 22.6 %, in_2 = 25.6 %). However, the majority of measures maintained reasonably balanced distributions without severe distortion. Among four dummy variables, two exhibited modest skew, but not at extreme levels: *inc* (≥$25,000 =28.1 %), and *lresd* (≥10years=26.1 %). Two land banking program variables, *part* (93.5 %) and *dlba* (79.3 %), exhibited high concentrations of zeros, indicating potential zero inflation. This distributional feature is theoretically expected, since these programs are designed to target only specific distressed properties. Moreover, because the WLSMV estimator treats such variables as categorical responses, the presence of zero inflation does not pose a direct bias risk, unlike in ML estimation that relies on distributional assumptions. The *part* and *dlba* variables, however, were recoded into dummy variables (0 vs. 1 +) to reduce the influence of excessive zeros and to facilitate model estimation and interpretation. Lastly, although not essential for WLSMV estimation, we checked multicollinearity and found minimal overlap among predictors, with an average VIF of 1.21, indicating no concerns.

Different types of goodness-of-fit indices support the fact that the model demonstrates a good fit (Gim and Park, 2025); for instance, RMSEA (= 0.053), SRMR (= 0.045), and CFI (= 0.989) all fall within the conventional thresholds for good model fit. Given the known sensitivity of the chi-square statistic to sample size—particularly when the sample exceeds 500—we report the relative chi-square (χ^2/df = 2.653), also known as the normed chi-square, which is less affected by sample size and falls within the acceptable range (typically < 3). Furthermore, to supplement the single-point RMSEA estimate, we provide its 90 % confidence interval (0.047–0.058). Since the upper bound remains below the 0.06 threshold, the model fit can be interpreted as good with respect to this index as well.

4. Results

4.1. Descriptive statistics

Among the programs evaluated, Side Lot had the highest average count, followed by Demolition, Own It Now, LB-owned properties, and Partner Sales, indicating that more than one Side Lot and Demolition activity typically appear around respondents on average. The mean scores for SC measures averaged around 3.44, indicating a moderate to somewhat high level of SC. Network-related items scored the highest (mean=3.60), followed by Support (=3.60), Place Attachment (=3.35), and Community Satisfaction, which scored relatively lower (=3.14). Inversely, survey respondents were somewhat dissatisfied with their neighborhood conditions, rating them slightly below neutral, with safety receiving relatively lower satisfaction scores compared to other aspects. Overall, the findings indicate that strong social ties persist even as residents report declining satisfaction with their physical and communal environments (Table 2).

4.2. WLSMV results

The outcomes of the structural and measurement model presented in Table 3 and Fig. 3 show the standardized path coefficient and squared multiple correlations (SMC) of endogenous variables. Most indicators in the measurement model were significant, with appropriate SMC values: 0.063 for SC and 0.633 for PNC, indicating moderate to strong explanatory power.

When observing the effect of exogenous variables on SC, place attachment (*pa_2*= 0.812) had the largest effect, suggesting it is the most influential indicator of SC followed by community satisfaction (*csf_2*=0.802) and support (*is_3*=0.678). Additionally, PNC is well explained by its observed indicators, with improvement (*improve*=0.838), safety (*safety*=0.466), home value (*hvalue*=0.812), and cleanliness (*clean*=0.806), highlighting its strong contribution to the construct.

In the structural model, among the five LB programs, Own-It-Now (*oin*=-0.125), Side Lot (*side*=-0.094), and DLBA owned (*dlba*=-0.087) showed negative associations with PNC ($p < 0.05$), suggesting these programs may lead to dissatisfaction or fail to improve perceptions toward the neighborhood effectively. Income (*income*=0.105) demonstrated a significant positive relationship ($p < 0.05$), while age and homeownership were not significant. The SMC for PNC was 0.063; although weak, it was considered acceptable as it is likely influenced by other possible covariates not fully included in the model. PNC exhibited a strong positive linkage with SC (=0.776, $p < 0.01$), indicating that better neighborhood conditions are connected to higher levels of SC. Among the LB programs, Side Lot (*side*=0.061) and Partner Sales (*part*=0.057) showed a relatively weak but significant positive association with SC, whereas Demolition (*dem*=-0.062) showed negative relationship ($p < 0.05$). Additionally, age (*age*=0.115, $p < 0.01$) and homeownership (*hown* =0.098, $p < 0.01$) displayed positive relationship with SC.

Table 4 presents the direct, indirect, and total effect on factors such as PNC (*NEI_CON*) and SC (*SC*). The indirect effects of LBs on SC were mostly insignificant, except for Own-it Now, Side-Lot program, and DLBA-owned (actually vacant properties). In the case of SC, only Own-It-Now program showed a statistically significant negative total effect (=-0.066, $p < 0.1$), combinig direct positive (=0.031) and indirect

Table 2
Descriptive Statistics (N = 598).

Construct	Var.	Descriptions		Mean	S.D.	min	Max
LB Programs	dem	Demolition		1.070	1.645	0	10
	oin	Own-It-Now		0.891	1.652	0	12
	side	Side Lot		1.135	1.894	0	15
Perceived Neighborhood Condition*	upkeep	Upkeep of homes and yards in the neighborhood		2.998	1.151	1	5
	improve	Neighborhood improvement		2.824	1.212	1	5
	safety	Safety (e.g., crime, arson, or vandalism) in the community		2.587	1.238	1	5
	hvalue	Home values in the neighborhood		2.737	1.215	1	5
	clean	Cleanliness of the neighborhood		2.768	1.274	1	5
Social Capital†	is_1	[Support] There are neighbors in this neighborhood who are willing to help me.		3.413	1.076	1	5
	is_2	[Support] I know neighbors whom I am willing to help when they need my help.		3.763	1.009	1	5
	is_3	[Support] I am willing to help the neighbors in the entire community.		3.612	1.003	1	5
	in_1	[Network] I have a neighbor whom I enjoy a daily conversation with.		3.487	1.092	1	5
	in_2	[Network] When walking around the neighborhood, I say hello to the neighbors.		3.709	1.107	1	5
	csf_1	[Community Satisfaction] There are some places to hang out with the neighbors.		3.067	1.173	1	5
	csf _2	[Community Satisfaction] The quality of living in this neighborhood has improved.		3.219	1.067	1	5
	pa_1	[Place attachment] I am willing to continue to live in this neighborhood.		3.440	1.198	1	5
	pa_2	[Place attachment] Living in this neighborhood makes me feel a sense of community.		3.269	1.116	1	5
		Descriptions	Codes	Descriptions		Freq.	%
LB Programs	part	Partner Sales	0	absence		559	93.5
			1	presence		39	6.5
	dlba	LB-owned	0	absence		474	79.3
			1	presence		124	20.7
Individual Characteristics	hown	Home ownership	0	Rent		285	47.7
			1	Own		313	52.3
	lresd	Length of residence	0	≤ 10 years		442	73.9
			1	> 10 years		156	26.1
	inc	Household Income	0	≤ $25,000		168	28.1
			1	> $25,000		430	71.9
	age	Age	0	< 40		277	46.3
			1	≥ 40		321	53.7

* 1 “not at all satisfied” to 5 “extremely satisfied”
† 1 “strongly disagree” to 5 “strongly agree”

Table 3
WLSMV Results.

				Standardized coef.		S.E.	p	SMC
Structural model	NEI_CON	←	dem	−0.011		0.018	0.791	0.063
		←	oin	−0.125	***	0.018	0.000	
		←	side	−0.094	**	0.015	0.023	
		←	part	0.001		0.118	0.987	
		←	dlba	−0.087	**	0.077	0.049	
		←	hown	0.033		0.067	0.483	
		←	lresd	0.000		0.075	0.997	
		←	inc	0.105	**	0.074	0.025	
		←	age	−0.038		0.063	0.384	
	SC	←	NEI_CON	0.776	***	0.039	0.000	0.633
		←	dem	−0.062	*	0.014	0.081	
		←	oin	0.031		0.012	0.324	
		←	side	0.061	*	0.012	0.075	
		←	part	0.057	*	0.079	0.061	
		←	dlba	0.050		0.057	0.163	
		←	hown	0.098	***	0.047	0.007	
		←	lresd	−0.059	*	0.051	0.091	
		←	inc	0.060		0.054	0.111	
		←	age	0.115	***	0.043	0.001	
Measurement model	upkeep	←	NEI_CON	0.708	†			0.502
	improve	←	NEI_CON	0.838	***	0.045	0.000	0.702
	safety	←	NEI_CON	0.766	***	0.043	0.000	0.587
	hvalue	←	NEI_CON	0.812	***	0.043	0.000	0.659
	clean	←	NEI_CON	0.806	***	0.037	0.000	0.650
	is_1	←	SC	0.640	†			0.410
	is_2	←	SC	0.660	***	0.040	0.000	0.436
	is_3	←	SC	0.678	***	0.046	0.000	0.459
	in_1	←	SC	0.692	***	0.046	0.000	0.478
	in_2	←	SC	0.644	***	0.051	0.000	0.415
	csf_1	←	SC	0.656	***	0.050	0.000	0.431
	csf_2	←	SC	0.802	***	0.054	0.000	0.643
	pa_1	←	SC	0.724	***	0.051	0.000	0.524
	pa_2	←	SC	0.812	***	0.054	0.000	0.659

[*] $p < 0.1$
[**] $p < 0.05$
[***] $p < 0.01$
[†] Fixed to 1 per factor in the measurement model

negative effects (=-0.097) This indicates that Own-It-Now program weakens SC both directly and indirectly. By contrast, for Side Lot, the direct effect was positive (=0.061), strengthening SC, but the indirect effect was negative (=-0.073) and larger in magnitude, resulting in a total effect that was negative (=-0.012) but not statistically significant.

5. Discussion

For RQ1, the study concluded that some LB programs have direct effects on SC, while others do not. The Side Lot and Partner sales programs were found to have positive direct effects on SC, whereas Demolition showed a negative direct effect. For RQ2, the mediation effects of perceived neighborhood conditions (PNC) varied across land bank programs. In particular, Own It Now showed an indirect negative effect on SC through PNC. In contrast, the Side Lot program demonstrated a modest positive direct effect on SC, but this was outweighed by a larger negative indirect effect through PNC, resulting in a contradictory overall mechanism and no significant total impact.

There is no specific evidence to fully explain this contradictory impact of the Side Lot program. The repurposing of abandoned land done mostly by the remaining residents in SC has been shown to strengthen and is likely to further reinforce existing residents' commitment to the neighborhood. Even residents who did not purchase a side lot through the program may observe the financial and emotional investment made by their neighbors. These visible signs of positive change and stewardship foster a collective sense of pride and stewardship, encouraging shared responsibility for the neighborhood. Over time, this mutual appreciation can cultivate a stronger sense of community and belonging, reinforcing social ties and membership within the neighborhood. Possibly, inhabitants who do not share borders due to the empty lot between them could be physically closer after vacant lots are turned into a neighbor's side yard, which promotes interpersonal relationships (Park and Rogers, 2015). This finding aligns with the observations made by Rigolon et al. (2021) through interviews in Chicago with individuals, where the physical improvement of vacant lots was found to stimulate conversations and re-establish neighboring and community interaction. Such actions can amplify place attachment, as individuals begin to associate the neighborhood with a shared identity and emotional connection (Nassauer and Raskin, 2014). However, its positive impacts on SC may be tempered by broader perceptions of neighborhood decline. Many of these lots, while physically improved, often fall short of community aesthetic expectations or remain isolated amid persistent blight. The efforts of residents to repurpose abandoned land still contribute to a subtle yet meaningful shift in how the neighborhood's investment could be valued. At the same time, the persistence of nearby deterioration or inconsistent outcomes of side lot development may undermine these intended social benefits, resulting in an overall contradictory effect on SC. Future research should further investigate how side lots are developed, the purposes for which they are used, and the extent to which they contribute to tangible neighborhood improvements.

However, the positive impacts of side lots on SC may be tempered by the program's structural characteristics and the broader context of neighborhood decline. Three key features of the Side Lot program may help explain these contradictory effects. First, at $100 per parcel, the Side Lot program is accessible to many residents, which explains why it accounts for the largest share of land bank transactions. From 2014–2020, side lots accounted for 67.4 % of all land bank sales (Detroit, 2020). However, this low purchase price means that many lots are acquired with little incentive or resources for significant

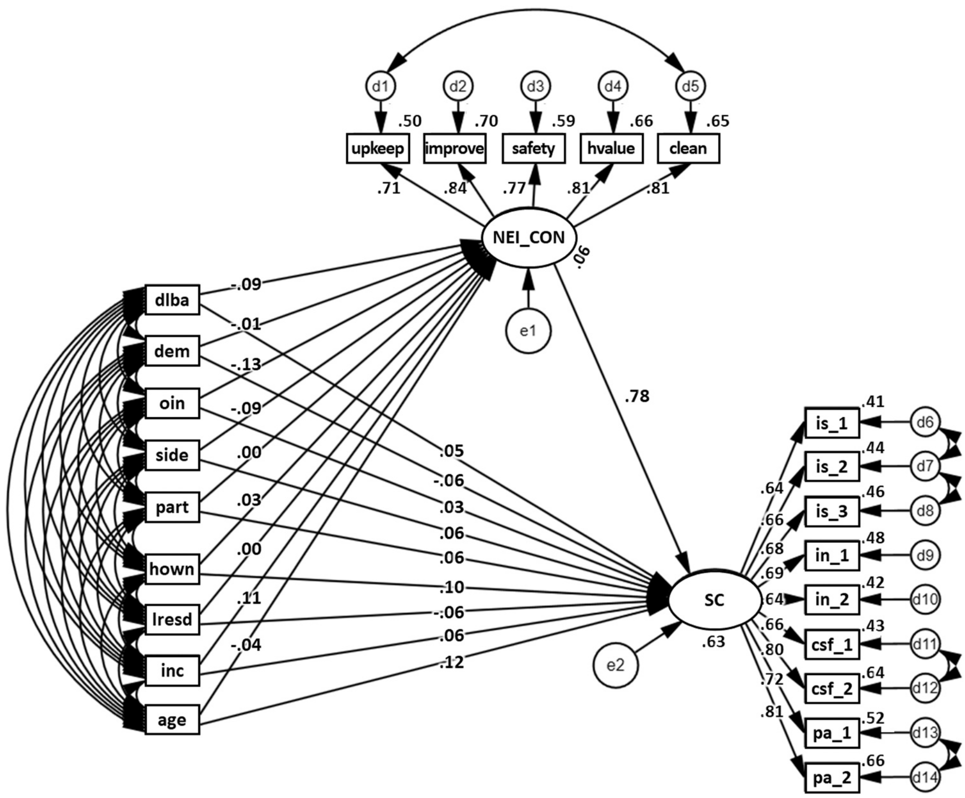

Fig. 3. Structural Equation Model & Results. Note: To reduce visual clutter, the covariance paths among observed variables were excluded from the main figure, as they did not aid interpretation.

Table 4
Direct, indirect, and total effects of LB variables on Perceived Neighborhood Condition and Social Capital.

Variables	Direct/Indirect/ Total	NEI_CON	SC
Demolition	Direct	−0.011	−0.062*
(dem)	Indirect		−0.008
	Total†	−0.011	−0.070
Own It Now	Direct	−0.125***	0.031
(oin)	Indirect		−0.097***
	Total†	−0.125	−0.066*
Side Lot	Direct	−0.094**	0.061*
(side)	Indirect		−0.073**
	Total†	−0.094	−0.012
Partner Sales	Direct	0.001	0.057*
(part)	Indirect		0.001
	Total†	0.001	0.058
LB-owned	Direct	−0.087**	0.050
(dlba)	Indirect		−0.067*
	Total†	−0.087	−0.017

* $p < 0.1$
** $p < 0.05$
*** $p < 0.01$
† Total = Direct + Indirect

reinvestment, such as landscaping, amenities, or community functions. Second, because side lots are typically small, adjacent parcels, improvements remain localized to the household level. Such changes often fail to meet broader aesthetic standards or shift collective perceptions of neighborhood decline. While individual residents may benefit from expanded property, these incremental improvements do not necessarily translate into visible neighborhood-wide revitalization. Third, when a program dominates land bank activity but delivers small, inconsistent outcomes, its collective contribution to SC may be diluted. Instead of signaling neighborhood revitalization, the prevalence of side lot development can normalize piecemeal or uneven land reuse, potentially reinforcing perceptions of persistent blight rather than transformative change. Therefore, future research should further investigate how side lots are developed, the purposes for which they are used, and the extent to which different types of improvements contribute to tangible neighborhood-level benefits beyond individual household gains.

Demolition led to a statistically significant decrease in SC. Past studies indicate that demolition is a controversial tool in vacancy response (Mallach, 2011). Sikanas (2015) state that demolition leads to a decreased sense of place and increased stress, depression, and anxiety, as people lose their memory of place. Prener et al. (2020) found that individuals in communities often develop an emotional attachment to homes scheduled for demolition, even when these properties have been abandoned or deteriorated for extended periods. This attachment may stem from the collective memories and neighborhood identity that such structures represent. Further, demolition alone fails to contribute to any substantial improvement in PNCs. Demolition also risks leaving behind vacant lots or barren spaces, especially when the properties cannot be sold or repurposed immediately. This is partially supported by findings from Yin and Silverman (2015) which revealed no significant impact of demolition on housing sales prices, as it fails to enhance the neighborhood's appeal, particularly if no immediate follow-up actions are undertaken after demolition (vom Hofe et al., 2019). Thus, demolition primarily affects SC directly, through the removal of place-based anchors, while indirect improvements via PNC are largely absent. Further research, however, on the impact of demolition on surrounding areas is necessary because the decisions about which properties to demolish often depend on specific local contexts. For instance, as Herbert (2018) noted, cities or LBs may selectively target properties for demolition in areas they perceive as more declined or strategically favorable. This suggests that the criteria guiding such decisions are complex and shaped by varying priorities and strategic objectives.

Third, the presence of Own-It-Now sold properties nearby had a noticeable total negative impact on SC, as the negative influence of PNC outweighed the direct positive impacts. For instance, while the compliance guidelines for Own-It-Now properties ensure basic habitability through functional kitchens, bathrooms, utilities, and a maintained exterior, these standards focus primarily on meeting minimum requirements rather than encouraging higher-level remodeling or substantial improvements (Detroit Land Bank Authority, 2024). Presently, there is insufficient data on the degree to which homes sold through the Own-It-Now program have been renovated. For example, a property sold for as little as $1000 is unlikely to have undergone extensive remodeling. Moreover, the timeline for renovations can vary significantly, with more comprehensive remodeling efforts potentially taking longer to complete. One survey respondent [Respondent #78] offers a remark that supports our potential explanation:

"People think they get a good deal with the land bank, but they really don't. They make you fix it up completely within six months. If you can't do that, the land bank will take it back."

This suggests that current program requirements may unintentionally hinder successful outcomes. To address this, policy improvements are needed to ensure that the transfer of vacant housing—often in poor condition—results in the supply of quality housing. Recommended improvements include extending renovation timelines, breaking the process into manageable steps, and providing adequate support to help purchasers complete the required upgrades.

Although some studies suggest that the arrival of newcomers can enhance social capital by introducing new networks, resources, and relationships (Rigolon et al., 2021), in our case, this effect appears modest in magnitude, likely constrained by residents' negative perceptions of newcomers as profit-seekers rather than community-minded participants. A possible explanation could be found in the open-ended question:

"*If someone chose to purchase a property in your neighborhood that is under the DLBA program, what do you think would be the reason for their purchase?*" While only nine respondents provided answers—limiting statistical generalizability—the responses most reflected profit-oriented intentions of new commers. These included:

"To refurbish and flip" [#119], "To flip it for a profit or tear it down" [#298], "To flip or rent out the property for profit" [#488], "Landlord obtaining rentals on a budget" [#327], and "Try and scam people to make even more money" [#414].

These responses suggest that residents may view them primarily as profit-seekers—such as flippers or landlords—rather than as community-minded participants. As a result, SC may not be formed greatly following such property transfers. As a result, any positive effects of newcomer inflows on SC are likely to remain modest. While the dynamics of SC are known to vary depending on the characteristics of incoming residents (Butler and Robson, 2001), the limited data in this study constrain our ability to draw firm conclusions about the relationship between land bank (LB) programs and gentrification, as well as the broader impact of gentrification on SC. Further research is needed to examine how the socio-economic profiles and motivations of incoming residents shape neighborhood dynamics and influence the formation, disruption, or transformation of SC during neighborhood revitalization or residential gentrification processes.

Fourth, the Partner Sales program showed a weak but statistically significant effect on SC. This finding aligns with existing literature demonstrating that community-based collaborative initiatives foster social benefits through collective engagement toward shared objectives, resulting in increased SC (Christensen et al., 2019; Glover et al., 2005). Nevertheless, to transform this program's weak association into a stronger one, support is needed to facilitate more rapid project progression. To illustrate, only 49 community partner sale lots and structures were sold around the respondents' homes, which only explains 3.3 % of the total Partner Sales in Detroit during the 3-year observation period. We investigated each partner-sale property with Google Street View (GSV) images to identify how these properties are utilized. Twenty of the 49 structures and lots were either vacant or severely neglected, and, apart from the nine spots where GSV images are not available after the sales date, 20 lots are currently being reused. The postponed transformation could be because different "hold durations" are allowed for sold for community projects. According to compliance requirements for the Own-It-Now and Auction programs, sold properties should be occupied within 60 days, while the DLBA allows a maximum 6-month hold for Partner Sales, if there is a reasonable purpose (Detroit Land Bank Authority, 2021). Furthermore, it often takes longer for community partners to repurpose the properties. For instance, the Georgia Street Community Garden Collective was initiated in November 2008 but became a non-profit organization in 2009 (Hahn, 2013). The project's implementation process was even more protracted; Google Street View images show that no activity had occurred by 2021, and even by 2022, only partial progress—roughly half—had been made (from Google Earth). By collecting more cumulative data from the Partner Sales program, expanding the target areas, comparing participants and non-participants, and interviewing community members, the impact of community Partner Sales on SC could be better understood.

Lastly, renters and low-income residents tended to have less SC than their counterparts. Indeed, several previous studies have suggested that individuals with low incomes often experience social mistrust, a lack of social membership, and disinvestment in SC (Groot et al., 2007; Kawachi et al., 1997). Lower-income individuals have been shown to less likely to participate in community activities due to long working hours and limited resources to invest in SC (Huete, 2016; McCabe, 2012). Moreover, it is well known that concerns regarding housing costs typically drive homeowners to participate in local decision-making, networking, and active participation in community activities in order to reap the benefits of having control over their home and community (Roskruge et al., 2013).

The findings from this study reveal that the impacts of LB programs on SC are highly program dependent. For instance, the Side Lot program exhibits a modest positive direct effect on SC, suggesting that when residents actively reoccupy and repurpose adjacent vacant lots, there is an enhancement in community ties and interpersonal support. However, this benefit is partially offset by a negative indirect effect mediated through PNCs, indicating that while residents may benefit directly from increased neighborly engagement, they also experience a decline in overall neighborhood satisfaction, possibly due to aesthetic or safety concerns. In contrast, demolition consistently shows a clear negative impact on both neighborhood conditions and SC, echoing previous findings that highlight the loss of neighborhood identity and cohesion following the removal of long-standing structures (Mallach, 2011; Sikanas, 2015). These mixed outcomes underscore the importance of contextual factors that were not fully explored in the current study. For example, historical disinvestment practices such as redlining and variations in local governance structures could significantly shape how LB programs influence community dynamics (Robinson and Woodin, 2024). Given such variability, our policy recommendations must be interpreted with caution. Rather than advocating for uniform interventions, policymakers should consider context-specific strategies. For instance, in neighborhoods where the negative impacts of demolition on SC are pronounced, complementary measures—such as post-demolition redevelopment plans and enhanced community engagement initiatives—may be necessary to mitigate adverse effects. Similarly, while the Side Lot program shows promise, its implementation should be accompanied by efforts to improve the broader physical environment to maximize its positive impact on SC.

6. Conclusion

The findings reveal that Land Bank programs influence SC not only through direct pathways but also indirectly via PNC. However, these effects were not consistent across all programs. For instance, Own-It-Now exerted small direct positive effects, but these were outweighed by negative indirect effects through PNC, resulting in a statistically significant total negative impact. Side Lot displayed a contradictory mechanism in which positive direct effects were offset by stronger negative indirect effects, leading to an insignificant total outcome, while Demolition showed a direct negative effect. To achieve this, the government should not only provide clear guidelines but also offer accessible tools such as loans, toolkits, or low-interest financing to support residents in meeting program requirements and neighborhood improvements. Additionally, public awareness campaigns—using word-of-mouth, direct outreach, educational materials, and incentives—are essential to boost resident participation in reoccupation programs. While VPs remain listed but unoccupied, they can contribute to neighborhood decline and potentially harm SC in the long term. To address this, strategic actions such as removing visible signs of neglect (e.g., graffiti, vandalism, litter) and implementing alternative security measures (e.g., lighting, fencing, surveillance) are critical. Regular maintenance audits led by active resident groups can further mitigate deterioration. These audits also provide opportunities to promote LB and government programs through signage, signaling that efforts are underway to manage VPs. Finally, it is essential to carefully assess reuse potential before pursuing demolition. A clear reuse plan, effectively communicated to the public, can reduce the time VPs remain vacant and minimize the negative impacts of demolition. These actions are particularly crucial in low-income neighborhoods and should be supported by LBs, local governments, and other relevant institutions.

While the findings of this study contribute to understanding and advancing LB programs, further research could enhance these insights by overcoming limitations of this study. While a quantitative model offers statistical rigor, it does not fully account for the emotional, symbolic, and contextual dimensions through which residents understand and respond to land bank interventions. Incorporating qualitative data—such as in-depth interviews, focus groups, or participant observation—could provide insight into how residents emotionally interpret land bank interventions, perceive their own agency, and construct the meanings behind their survey responses. This limitation has been noted as a suggested direction for future research. In addition, because SC formation entails both short-term and long-term processes, it is important to consider not only the strength of ties within relatively homogeneous groups such as neighborhoods or close-knit communities (i.e., bonding SC, which is the focus of this study), but also broader and more diverse connections that link different social groups (bridging SC), as well as relationships that connect individuals to institutions and power structures. Thus, future studies should consider employing repeated survey waves over time. A longitudinal design would enable pre- and post-intervention analysis, offering a more comprehensive understanding of how SC evolves in response to land bank initiatives and related urban interventions. Additionally, the potential influence of the COVID-19 pandemic on SC is valid, as the survey was conducted in June 2021. To draw conclusions that extend beyond the pandemic period, longitudinal data would be instrumental in capturing the dynamics of SC in the post-pandemic context.

In future research, it would be valuable to integrate both subjective and objective indicators of neighborhood condition to more precisely examine how spatial context moderates the effects of land bank interventions. Given that our current study used individual-level units of analysis defined by respondent-centered buffers, obtaining comprehensive objective neighborhood conditions at this granular spatial scale was not feasible. Should data be collected at a higher spatial unit of analysis, such as census tracts or planning districts, the modeling approach would necessarily differ. For example, applying a Multilevel Structural Equation Model (MSEM) would require not only a sufficient number of groups, but also an adequate number of individuals within each group—typically 10–30 cases per unit (Kush et al., 2022)—to ensure statistical stability. Census tracts, which are commonly used proxies for neighborhood-level characteristics and contain the most comprehensive spatial data, present a feasible option. As of 2024, the City of Detroit contains approximately 280 census tracts, which implies that a sample of at least 2800 individuals would be necessary for a statistically robust MSEM. At the same time, it is important to acknowledge that SC is a multidimensional construct that extends beyond proposed measures in this study. Our current operationalization, while robust in capturing residents' immediate perceptions and interpersonal support, may underrepresent the formal and broader bridging dimensions (e.g., residents' participation in local governance, membership in civic organizations, and engagement with community institutions) that are critical for understanding the full spectrum of SC.

Future studies may also further investigate the possibility of reverse or reciprocal causality between land bank activities and neighborhood conditions. While the current study assumes that objective interventions (e.g., land bank programs) influence subjective perceptions (e.g., neighborhood conditions), it is equally plausible that neighborhoods already perceived as distressed may attract more listings, rather than land bank activity causing negative perceptions. This suggests that the possibility of bidirectional or simultaneous causality merits further empirical exploration.

Relatedly, as LB programs and similar vacancy responses have diversified across the U.S. and internationally as many cities have experienced population decline that led to increased vacancy and the need for legitimate institutional responses. In developed countries, declining birth rates—alongside inter-city migration—are key risk factors contributing to the intensification of future vacancy issues (Park et al., 2021). For example, the point at which the replacement fertility rate of 2.0, the threshold required to sustain a generation's population, fell below this level occurred in 1975 for Japan, 1984 for South Korea, and 2010 for the U.S; as of 2023, the total fertility rates stood at 1.62 in the U.S., 1.21 in Japan, and a record-low 0.72 in South Korea (Romei, 2024). Although South Korea reached below-replacement fertility 13 years later than Japan, the rate of decline has been considerably steeper (Statistics Korea, 2024). As this study has illustrated, population decline not only leads to economic hardship but also contributes to the erosion of SC. In response to the rapid demographic shifts that have driven a surge in VPs, a broader and more proactive range of policy interventions will soon be needed in many countries. Given that the drivers and characteristics of vacancy differ across national and local contexts—including those in the Global South, it is essential to conduct comparative studies across different contexts and programs to enhance the broader applicability of findings beyond the Detroit case.

CRediT authorship contribution statement

Yunmi Park: Writing – original draft, Investigation, Funding acquisition, Conceptualization. **Hyungchul Chung:** Writing – review & editing, Methodology, Data curation. **Jung-Eun Lee:** Visualization, Validation, Data curation. **Tae-Hyoung Tommy Gim:** Methodology, Formal analysis, Conceptualization. **Newman Galen Daniel:** Writing – review & editing, Validation.

Declaration of Competing Interest

I, Yunmi Park, hereby declare that I, including other authors, have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this document. I confirm that this declaration is accurate to the best of my knowledge and that I have adhered to ethical guidelines in conducting and reporting this work.

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Data availability

Data will be made available on request.

References

Adler, P.S., Kwon, S.-W., 2002. Social capital: prospects for a new concept. Acad. Manag. Rev. 27 (1), 17–40.

Agheyisi, J.E., 2025. Temporary use of urban vacant spaces: a pro-poor land use strategy in the Global South. J. Urban Aff. 47 (3), 978–1002.

Akiya Banks, 2024. About. Retrieved Jan. 21. 2025 from https://www.akiyabanks.com/about-5.

Alm, J., Hodge, T.R., Sands, G., Skidmore, M., 2014. Detroit property tax delinquency: Social contract in crisis. Lincoln Institute of Land Policy.

Anderson, E.C., Minor, E.S., 2017. Vacant lots: an underexplored resource for ecological and social benefits in cities. Urban For. Urban Green. 21, 146–152.

Anton, C.E., Lawrence, C., 2014. Home is where the heart is: the effect of place of residence on place attachment and community participation. J. Environ. Psychol. 40, 451–461.

de Araujo Larangeira, A., 2003. Vacant land in Latin America: challenges and opportunities. Linc. Inst. Land Policy 39 (Cambridge).

Baker, R., 2020. Racial capitalism and a tentative commons: Urban farming and claims to space in post-bankruptcy Detroit. Commoning the City. Routledge, pp. 25–36.

Bearre, C., 1976. Farm-a-Lot: Detroiters turn vacant lots into gardens. Ann Arbor Sun.

Blanco, E.C., 2020. As Michigan Makes Progress on Vacant Homes, Detroit's Vacancies Have Skyrocketed. City. ⟨https://nextcity.org/urbanist-news/michigan-makes-progress-on-vacant-homes-detroits-vacancies-have-skyrocketed⟩.

Bollwahn, B., 2019. Property Disposition Matters: The Current Status of Land Bank Programs in the United States..

Bourdieu, P., 1986. The forms of capital. Handbook of theory and research for the sociology of education. JG Richardson 241 (258), 19 (New York, Greenwood).

Boyce, W.F., Davies, D., Gallupe, O., Shelley, D., 2008. Adolescent risk taking, neighborhood social capital, and health. J. Adolesc. Health 43 (3), 246–252.

Brennan, S.P., 1980. Shaping urban futures; an examination of land banking. Urban Growth Urban Form. Carlet. Univ.

Brown, G., Raymond, C., 2007. The relationship between place attachment and landscape values: toward mapping place attachment. Appl. Geogr. 27 (2), 89–111.

Butler, T., Robson, G., 2001. Social capital, gentrification and neighbourhood change in London: a comparison of three south London neighbourhoods. Urban Stud. 38 (12), 2145–2162.

Caughy, M.O.B., O'campo, P.J., 2006. Neighborhood poverty, social capital, and the cognitive development of African American preschoolers. Am. J. Community Psychol. 37 (1-2), 141.

Center For Community Progress, 2019. ational Map of Land Banks & Land Banking Programs. ⟨https://www.communityprogress.net/land-bank-map-pages-447.php⟩.

Center For Community Progress, 2024a. National Land Bank Map. ⟨https://communityprogress.org/resources/land-banks/national-land-bank-map/⟩.

Center For Community Progress. (2024b, Oct. 28. 2024). *Why Land Banks Matter*. Retrieved Jan. 30. 2025 from ⟨https://communityprogress.org/blog/why-land-banks-matter/⟩.

Christensen, S., Malberg Dyg, P., Allenberg, K., 2019. Urban community gardening, social capital, and" integration"–a mixed method exploration of urban" integration-gardening" in Copenhagen, Denmark. Local Environ. 24 (3), 231–248.

Clark, W.A.V., Onaka, J.L., 1983. Life cycle and housing adjustment as explanations of residential mobility. Urban Stud. 20 (1), 47–57.

Coleman, J.S., 1988. Social capital in the creation of human capital. Am. J. Sociol. 94, S95–S120.

Cook County Land Bank Authority, 2023. Land bank authority economic impact one year snapshot.

Curley, A.M., 2010. Relocating the poor: social capital and neighborhood resources. J. Urban Aff. 32 (1), 79–103.

Dang, T. (2024, 16 July 2024). *Top 6 Akiya Banks For Foreigners To Use In 2024*. Retrieved Jan. 22. 2025 from ⟨https://mailmate.jp/blog/akiya-banks-for-foreigners⟩.

Dassopoulos, A., Batson, C.D., Futrell, R., Brents, B.G., 2012. Neighborhood connections, physical disorder, and neighborhood satisfaction in Las Vegas. Urban Aff. Rev. 48 (4), 571–600.

Detroit. (2021). *Neighborhood Stabilization Program*. Retrieved 01/25/2022 from ⟨https://detroitmi.gov/departments/housing-and-revitalization-department/hud-programs-and-information/neighborhood-stabilization-program⟩.

Detroit, T.C. o, 2020. DBLA Sold. ⟨https://data.detroitmi.gov⟩.

Detroit Land Bank Authority, 2020. City Council Quarterly Report – April 2020. ⟨https://buildingdetroit.org/dlba-reports⟩.

Detroit Land Bank Authority, 2021. Detroit Land Bank Programs. ⟨https://buildingdetroit.org/programs⟩.

Detroit Land Bank Authority, 2024. Compliance. ⟨https://buildingdetroit.org/programs⟩.

Díaz, G.I., Nahuelhual, L., Echeverría, C., Marín, S., 2011. Drivers of land abandonment in Southern Chile and implications for landscape planning. Landsc. Urban Plan. 99 (3-4), 207–217.

Firth, C., Maye, D., Pearson, D., 2011. Developing "community" in community gardens. Local Environ. 16 (6), 555–568.

Flora, D.B., Curran, P.J., 2004. An empirical evaluation of alternative methods of estimation for confirmatory factor analysis with ordinal data. Psychol. Methods 9 (4), 466.

Stephen P. Gasteyer, R.J., Vicki C. Krueger (2015). Land banks: investing in communities, banking on revitalizatio. ⟨https://realtorparty.realtor/wp-content/uploads/2017/12/State-Local-issues-Land-Banks.pdf⟩.

Ghasri, M., Rashidi, T., Auld, J., 2023. Determinants of residential mobility: an adaptive retrospective survey method. Transp. Lett. 15 (2), 129–141.

Gim, T.H.T., Park, Y., 2025. Influences of COVID-19 pandemic on travel patterns across past, present, and future. Int. J. Urban Sci. 29 (2), 405–429.

Glover, T.D., Parry, D.C., Shinew, K.J., 2005. Building relationships, accessing resources: Mobilizing social capital in community garden contexts. J. Leis. Res. 37 (4), 450–474.

Graziani, K., 2024. The road ahead for land banks. C. F. C. Prog.

Groot, W., Van Den Brink, H.M., Van Praag, B., 2007. The compensating income variation of social capital. Soc. Indic. Res. 82 (2), 189–207.

Haase, D., 2008. Urban ecology of shrinking cities: an unrecognized opportunity? Nat. Cult. 3 (1), 1–8.

Hadavi, S., Rigolon, A., Gobster, P.H., Stewart, W.P., 2021. Resident-led vacant lot greening and crime: Do ownership and visual condition-care matter? Landsc. Urban Plan. 211, 104096.

Hahn, K., 2013. Georgia Street Community Collective – Part of Detroit's east side food system. (December 24, 2013). https://www.canr.msu.edu/news/georgia_street_community_collective_part_of_detroits_east_side_food_system.

Halpern, D., 2005. Social capital. Polity.

Hamilton County Land Bank, 2023. Land Bankung Programs. Retrieved 2023.07.01 from. ⟨https://www.hamiltoncountylandbank.org/how-we-help/our-programs/⟩.

Harpham, T., 2008. The measurement of community social capital through surveys. Social capital and health. Springer, pp. 51–62.

Heckert, M., 2015. A spatial difference-in-differences approach to studying the effect of greening vacant land on property values. Cityscape 17 (1), 51–60.

Heim Lafrombois, M.E., Park, Y., 2021. The uneven shrinking city: neighborhood demographic change and creative class planning in Birmingham, Alabama. Urban Geogr. 1–25.

Herbert, C.W., 2018. Like a good neighbor, squatters are there: property and neighborhood stability in the context of urban decline. City Community 17 (1), 236–258. https://doi.org/10.1111/cico.12275.

Hesari, E., Moosavy, S.M., Rohani, A., Besharati Kivi, S., Ghafourian, M., Saleh Sedgh Pour, B., 2020. Investigation the relationship between place attachment and community participation in residential areas: a structural equation modelling approach. Soc. Indic. Res. 151, 921–941.

Hong, C., 2018. The spatial spillover impact of land bank properties on nearby home sale values in Cleveland. OH Clevel. State Univ.

Huete, P.L., 2016. Social Capital in Low-Income Communities: Can It Reduce Poverty?.

Jones, C., Shen, J., 2014. Neighborhood social capital, neighborhood disadvantage, and change of neighborhood as predictors of school readiness. Urban Stud. Res. 2014.

Kawachi, I., Kennedy, B.P., Lochner, K., Prothrow-Stith, D., 1997. Social capital, income inequality, and mortality. Am. J. Public Health 87 (9), 1491–1498.

Kelly Jr,, J.J., 2014. Sustaining neighborhoods of choice: from land bank (ing) to Land Trust (ing). Washburn LJ 54, 613.

Kim, G., 2016b. The public value of urban vacant land: social responses and ecological value. Sustainability 8 (5), 486.

Kim, G., 2016a. Assessing urban forest structure, ecosystem services, and economic benefits on vacant land. Sustainability 8 (7), 679.

Kingsley, Y.J., Townsend, M., 2006. Dig in'to social capital: community gardens as mechanisms for growing urban social connectedness. Urban Policy Res. 24 (4), 525–537.

Kingston, S., Mitchell, R., Florin, P., Stevenson, J., 1999. Sense of community in neighborhoods as a multi-level construct. J. Community Psychol. 27 (6), 681–694.

Kleinhans, R., 2009. Does social capital affect residents' propensity to move from restructured neighbourhoods? Hous. Stud. 24 (5), 629–651.

Koay, W.I., Dillon, D., 2020. Community gardening: stress, well-being, and resilience potentials. Int. J. Environ. Res. Public Health 17 (18), 6740.

Kondo, M., Hohl, B., Han, S., Branas, C., 2016. Effects of greening and community reuse of vacant lots on crime. Urban Stud. 53 (15), 3279–3295.

Kondo, M., Degli Esposti, M., Jay, J., Morrison, C.N., Freisthler, B., Jones, C., Yang, J., Chisolm, D., Branas, C., Hohl, B., 2022. Changes in crime surrounding an urban home renovation and rebuild programme. Urban Stud. 59 (5), 1011–1030.

Kush, J.M., Konold, T.R., Bradshaw, C.P., 2022. The sampling ratio in multilevel structural equation models: Considerations to inform study design. Educ. Psychol. Meas. 82 (3), 409–443.

Kwok, R., 2018. Accidental urban oases. Proc. Natl. Acad. Sci. 115 (19), 4800–4804.

Lang, R.E., Hornburg, S.P., 1998. What is social capital and why is it important to public policy? Hous. Policy Debate 9 (1), 1–16.

Lee, J.-E., Park, Y., Newman, G.D., 2023. Twenty years of research on shrinking cities: a focus on keywords and authors. Landsc. Res. 1–16.

Lee, J.-E., Chung, H., Park, Y., 2025. From blight to boon: the impact of land banking programs on adjacent property values in Detroit, US. Cities 165, 106112. https://doi.org/10.1016/j.cities.2025.106112.

Lee, S.M., Conway, T.L., Frank, L.D., Saelens, B.E., Cain, K.L., Sallis, J.F., 2017. The relation of perceived and objective environment attributes to neighborhood satisfaction. Environ. Behav. 49 (2), 136–160.

Lelieveldt, H., 2004. Helping citizens help themselves: Neighborhood improvement programs and the impact of social networks, trust, and norms on neighborhood-oriented forms of participation. Urban Aff. Rev. 39 (5), 531–551.

Leonard, B., 2010. Revitalizing Foreclosed Properties with Land Banks. DIANE Publishing.

Leviten-Reid, C., Matthew, R.A., 2018. Housing tenure and neighbourhood social capital. Hous. Theory Soc. 35 (3), 300–328.

Leyden, K.M., 2003. Social capital and the built environment: the importance of walkable neighborhoods. Am. J. Public Health 93 (9), 1546–1551.
Lochner, K., Kawachi, I., Kennedy, B.P., 1999. Social capital: a guide to its measurement. Health Place 5 (4), 259–270. https://doi.org/10.1016/S1353-8292(99)00016-7.
Mallach, A., 2011. Demolition and preservation in shrinking US industrial cities. Build. Res. Inf. 39 (4), 380–394.
McCabe, B.J., 2012. Homeownership and social trust in neighbors. City Community 11 (4), 389–408.
Mead, G.H., 2015. Mind, self & society. University of Chicago press.
Meehan, J., Bryde, D.J., 2014. Procuring sustainably in social housing: The role of social capital. J. Purch. Supply Manag. 20 (2), 74–81.
Mischiu, D.E., 2019. Banking on land: A critical review of land banking in the United States.
Mohnen, S.M., Groenewegen, P.P., Völker, B., Flap, H., 2011. Neighborhood social capital and individual health. Soc. Sci. Med. 72 (5), 660–667.
Mouratidis, K., Yiannakou, A., 2022. What makes cities livable? Determinants of neighborhood satisfaction and neighborhood happiness in different contexts. Land Use Policy 112, 105855.
Mullenbach, L.E., Larson, L.R., Floyd, M.F., Marquet, O., Huang, J.-H., Alberico, C., Ogletree, S.S., Hipp, J.A., 2022. Cultivating social capital in diverse, low-income neighborhoods: the value of parks for parents with young children. Landsc. Urban Plan. 219, 104313.
Murugappan, M.N., King-Kallimanis, B.L., Mangir, C., Howie, L., Bhatnagar, V., Beaver, J.A., Kluetz, P.G., 2022. Floor and ceiling effects in the EORTC QLQ-C30 Physical Functioning subscale among patients with advanced or metastatic breast cancer. Cancer 128 (4), 808–818.
Nassauer, J.I., Raskin, J., 2014. Urban vacancy and land use legacies: a frontier for urban ecological research, design, and planning. Landsc. Urban Plan. 125, 245–253.
Newman, G., Hollander, J.B., Lee, J., Gu, D., Kim, B., Lee, R.J., Horney, J.A., Bearfield, D., Li, Y., 2018. Smarter shrinkage: a neighborhood-scaled rightsizing strategy based on land use dynamics. J. Geovisualization Spat. Anal. 2 (2), 11.
Niemesh, G., Jones-Farmer, L.A., Hart, J., Holmes, W., & Soundappan, N. (2019). The Impact of Land Bank Demolitions on Property Values.
Nolan, E.P., 2017. Vacant land conversion in Detroit, Michigan: A spatial analysis of neighborhood stabilization and communal access.
Nussbeck, F.W., Eid, M., Lischetzke, T., 2006. Analysing multitrait–multimethod data with structural equation models for ordinal variables applying the WLSMV estimator: What sample size is needed for valid results? Br. J. Math. Stat. Psychol. 59 (1), 195–213.
Park, S., Lee, J., Park, Y., 2022. Analysis of residential satisfaction changes by the land bank program using text mining. Sustainability 14 (18), 11707.
Park, Y., LaFrombois, M.E.H., 2019. Planning for growth in depopulating cities: An analysis of population projections and population change in depopulating and populating US cities. Cities 90, 237–248.
Park, Y., Rogers, G.O., 2015. Neighborhood planning theory, guidelines, and research: Can area, population, and boundary guide conceptual framing? J. Plan. Lit. 30 (1), 18–36.
Park, Y., Newman, G.D., Lee, J.-E., Lee, S., 2021. Identifying and comparing vacant housing determinants across South Korean cities. Appl. Geogr. 136, 102566.
Parker, E.A., Lichtenstein, R.L., Schulz, A.J., Israel, B.A., Schork, M.A., Steinman, K.J., James, S.A., 2001. Disentangling measures of individual perceptions of community social dynamics: results of a community survey. Health Educ. Behav. 28 (4), 462–486.
Perkins, D.D., Long, D.A., 2002. Neighborhood sense of community and social capital: A multi-level analysis. In Psychological sense of community: Research, applications, and implications. Springer, pp. 291–318.
Philadelphia Redevelopment Authority. (2010). Vacant land management in Philadelphia: The costs of the current system and the benefits of reform: Prepared by Penn Institute for Urban Research, Econsult Corporation.
Power, A., 2008. Does demolition or refurbishment of old and inefficient homes help to increase our environmental, social and economic viability? Energy Policy 36 (12), 4487–4501.
Power, A., 2010. Housing and sustainability: demolition or refurbishment? Proc. Inst. Civ. Eng. Urban Des. Plan. 163 (4), 205–216.
Prener, C.G., Braswell, T.H., Monti, D.J., 2020. St. Louis's "urban prairie": Vacant land and the potential for revitalization. J. Urban Aff. 42 (3), 371–389.
Putnam, R.D., 1995. Bowling alone: Americas declining social capital. J. Democr. 6 (1), 65–78.
Putnam, R.D., 2000. Bowling alone: America's declining social capital. In Culture and politics. Springer, pp. 223–234.
Reid, C.K., 2004. Achieving the American dream? A longitudinal analysis of the homeownership experiences of low-income families. Univ. Wash.
Rigolon, A., Banerjee, D., Gobster, P., Hadavi, S., Stewart, W., 2021. Transferring Vacant Lots to Private Ownership Improves Care and Empowers Residents: Evidence From Chicago. J. Am. Plan. Assoc. 87 (4), 570–584.
Robinson, S., Woodin, B., 2024. Michigan State Land Bank Authority Holdings, Historical Redlining, and Social Equity. J. Sustain. Real. Estate 16 (1), 2306708.
Rogers, G.O., Park, Y., 2018. Rapid Development, Build-Out Ratio and Subsequent Neighborhood Turnover. Sustainability 10 (5), 1339.
Rogers, S.H., Halstead, J.M., Gardner, K.H., Carlson, C.H., 2011. Examining walkability and social capital as indicators of quality of life at the municipal and neighborhood scales. Appl. Res. Qual. Life 6 (2), 201–213.
Romei, V., 2024. Birth rates in rich countries halve to hit record low. Financial. ⟨https://www.ft.com/content/f0d2a5a7-e5ef-4044-8380-ff690b609a5a⟩.
Roskruge, M., Grimes, A., McCann, P., Poot, J., 2013. Homeownership, social capital and satisfaction with local government. Urban Stud. 50 (12), 2517–2534.
Roy, A., 2011. Slumdog cities: Rethinking subaltern urbanism. Int. J. Urban Reg. Res. 35 (2), 223–238.
Ryu, H., Lee, J.S., Lee, S.-y, 2018. Participatory neighborhood revitalization effects on social capital: Evidence from community building projects in Seoul. J. Urban Plan. Dev. 144 (1), 04017025.
Saegert, S., Winkel, G., 2004. Crime, social capital, and community participation. Am. J. Community Psychol. 34 (3), 219–233.
Sampson, R.J., Raudenbush, S.W., Earls, F., 1997. Neighborhoods and violent crime: A multilevel study of collective efficacy. Science 277 (5328), 918–924.
Sands, G., & Skidmore, M. (2015). *Detroit and the Property Tax (Policy Focus Repor*, Issue. ⟨https://www.lincolninst.edu/sites/default/files/pubfiles/detroit-and-the-property-tax-full_0.pdf⟩.
Schieman, S., 2005. Residential stability and the social impact of neighborhood disadvantage: A study of-gender and race-contingent effects. Soc. Forces 83 (3), 1031–1064. https://doi.org/10.2307/3598268.
Schilling, J., Logan, J., 2008. Greening the rust belt: A green infrastructure model for right sizing America's shrinking cities. J. Am. Plan. Assoc. 74 (4), 451–466.
Schukoske, J.E., 1999. Community development through gardening: State and local policies transforming urban open space. NYUJ Legis. Pub. Pol. 'Y. 3, 351.
Sergeant, J.F., Ekerdt, D.J., 2008. Motives for residential mobility in later life: Post-move perspectives of elders and family members. Int. J. Aging Hum. Dev. 66 (2), 131–154.
Sikanas, S., 2015. Landscape of Loss: A Community Sensitive Approach to the Demolition in Detroit.
Simmel, G., 2009. The intersection of social circles. SIMMEL, G. Sociology: inquiries into the construction of social forms. Brill, *Boston*.
Skinner, C., Watson, V., 2017. The informal economy in cities of the global south: Challenges to the planning lexicon. Routledge Companion Plan. Glob. South 140–152.
Small, Z., Minner, J.S., 2024. Do land banks mean progress toward socially equitable urban development? Observations from New York State. Urban Aff. Rev. 60 (1), 272–303.
Statistics Korea, 2024. Population Status and Projections for the World and Korea Reflecting the 2022 Future Population Projections [Press Release]. ⟨https://kostat.go.kr⟩.
Stern, M., Lester, T.W., 2021. Does local ownership of vacant land reduce crime? an assessment of Chicago's large lots program. J. Am. Plan. Assoc. 87 (1), 73–84.
Stewart, W.P., Gobster, P.H., Rigolon, A., Strauser, J., Williams, D.A., van Riper, C.J., 2019. Resident-led beautification of vacant lots that connects place to community. Landsc. Urban Plan. 185, 200–209. https://doi.org/10.1016/j.landurbplan.2019.02.011.
Tappendorf, J.A., Denzin, B.O., 2010. Turning vacant properties into community assets through land banking. Urb. Law. 43, 801.
Temkin, K., Rohe, W.M., 1998. Social capital and neighborhood stability: An empirical investigation. Hous. Policy Debate 9 (1), 61–88.
U.S. Department of Health & Human Services, 2021. 2021 Poverty Guidelines, 2025/08/27. U.S. Department of Health & Human Services. ⟨https://aspe.hhs.gov/2021-poverty-guidelines⟩, 2025/08/27.
US Census Bureau. (2020). *Detroit city, Michigan; Michigan*.
Versey, H.S., 2018. A tale of two Harlems: Gentrification, social capital, and implications for aging in place. Soc. Sci. Med. 214, 1–11.
Veršinskas, T., Hartvigsen, M., & Gorgan, M. (2022). *European good practices on land banking: FAO study and recommendations*. Food & Agriculture Org.
vom Hofe, R., Parent, O., Grabill, M., 2019. What to do with vacant and abandoned residential structures? The effects of teardowns and rehabilitations on nearby properties. J. Reg. Sci. 59 (2), 228–249.
Whitaker, S., Fitzpatrick IV, T.J., 2016. Land bank 2.0: An empirical evaluation. J. Reg. Sci. 56 (1), 156–175.
Wiechmann, T., Wolff, M., 2013. Urban shrinkage in a spatial perspective–operationalization of shrinking cities in Europe 1990–2010. AESOPACSP Jt. Congr.
Wood, L., Shannon, T., Bulsara, M., Pikora, T., McCormack, G., Giles-Corti, B., 2008. The anatomy of the safe and social suburb: an exploratory study of the built environment, social capital and residents' perceptions of safety. Health Place 14 (1), 15–31.
Woolcock, M., 1998. Social capital and economic development: toward a theoretical synthesis and policy framework. Theory Soc. 27 (2), 151–208.
Woolley, M.E., Grogan-Kaylor, A., Gilster, M.E., Karb, R.A., Gant, L.M., Reischl, T.M., Alaimo, K., 2008. Neighborhood social capital, poor physical conditions, and school achievement. Child. Sch. 30 (3), 133–145.
Wu, X., Tan, X., 2023. Is the gated community really safe? An empirical analysis based on communities with varying degrees of closure. Int. J. Urban Sci. 27 (4), 725–745.
Yin, L., Silverman, R.M., 2015. Housing abandonment and demolition: exploring the use of micro-level and multi-year models. ISPRS Int. J. GeoInf. 4 (3), 1184–1200.